

Solution	Marks	
SECTION A 1. (a) L.H.S. = $1 + \frac{2 \sin 2x - \cos 3x}{\cos x}$ $= \frac{\cos x + 4 \sin x \cos x - (4 \cos^3 x - 3 \cos x)}{\cos x}$ $= \frac{4 \cos x + 4 \sin x \cos x - 4 \cos^3 x}{\cos x}$ $= 4 + 4 \sin x - 4 \cos^2 x$ $= 4 + 4 \sin x - 4(1 - \sin^2 x)$ $= 4 \sin x + 4 \sin^2 x = 4 \sin x(1 + \sin x) = \text{R.H.S.}$ (b) $2 + \frac{2 \sin 2\theta - \cos 3\theta}{\cos \theta} = 0 \implies 1 + \left(1 + \frac{2 \sin 2\theta - \cos 3\theta}{\cos \theta}\right) = 0$ Using (a): $1 + 4 \sin \theta(1 + \sin \theta) = 0$ $4 \sin^2 \theta + 4 \sin \theta + 1 = 0 \implies (2 \sin \theta + 1)^2 = 0$ $\sin \theta = -\frac{1}{2}$ Since $-\pi \leq \theta \leq \pi$, $\theta = -\frac{\pi}{6}$ or $-\frac{5\pi}{6}$	1M 1M 1A 1M 1M 1A	

Solution	Marks	
2. (a) L.H.S. = $(\cos A + \cos B) + (\cos C + \cos(A + B + C))$ $= 2 \cos \frac{A+B}{2} \cos \frac{A-B}{2} + 2 \cos \frac{A+B+2C}{2} \cos \frac{A+B}{2}$ $= 2 \cos \frac{A+B}{2} \left[\cos \frac{A-B}{2} + \cos \frac{A+B+2C}{2} \right]$ $= 2 \cos \frac{A+B}{2} \cdot 2 \cos \left(\frac{\frac{A-B}{2} + \frac{A+B+2C}{2}}{2} \right) \cos \left(\frac{\frac{A-B}{2} - \frac{A+B+2C}{2}}{2} \right)$ $= 4 \cos \frac{A+B}{2} \cos \frac{A+C}{2} \cos \left(-\frac{B+C}{2} \right) = 4 \cos \frac{A+B}{2} \cos \frac{B+C}{2} \cos \frac{C+A}{2} = \text{R.H.S.}$ (b) Let $A = x, B = 3x, C = 5x \implies A + B + C = 9x$ $\cos x + \cos 3x + \cos 5x + \cos 9x = 0$ Using (a): $4 \cos \frac{x+3x}{2} \cos \frac{3x+5x}{2} \cos \frac{5x+x}{2} = 0$ $4 \cos 2x \cos 4x \cos 3x = 0 \implies \cos 2x = 0, \cos 3x = 0, \text{ or } \cos 4x = 0$ For $-\frac{\pi}{3} < x < \frac{\pi}{6}$: If $\cos 2x = 0 \implies 2x = -\frac{\pi}{2} \implies x = -\frac{\pi}{4}$ If $\cos 3x = 0 \implies 3x = -\frac{\pi}{2} \implies x = -\frac{\pi}{6}$ If $\cos 4x = 0 \implies 4x = -\frac{\pi}{2} \text{ or } -\frac{3\pi}{2} \implies x = -\frac{\pi}{8} \text{ or } -\frac{3\pi}{8} \text{ (out of range)}$ $\therefore x = -\frac{\pi}{4}, -\frac{\pi}{6}, -\frac{\pi}{8}$	1M 1M 1A 1M 1A	

Solution	Marks	
3. (a) R.H.S. = $\cot x - 4 \cot 4x$ $= \frac{\cos x}{\sin x} - \frac{4 \cos 4x}{\sin 4x}$ $= \frac{\cos x}{\sin x} - \frac{\cos 4x}{\sin x \cos x \cos 2x}$ $= \frac{\cos^2 x \cos 2x - \cos 4x}{\sin x \cos x \cos 2x}$ $= \frac{\frac{1+\cos 2x}{2} \cdot \cos 2x - (2 \cos^2 2x - 1)}{\frac{1}{2} \sin 2x \cos 2x}$ $= \frac{\cos 2x - 3 \cos^2 2x + 2}{\sin 2x \cos 2x}$ $= \frac{(1 - \cos 2x)(3 \cos 2x + 2)}{\sin 2x \cos 2x}$ $= \frac{(2 \sin^2 x)(3 \cos 2x + 2)}{(2 \sin x \cos x) \cos 2x}$ $= \frac{\sin x(3 \cos 2x + 2)}{\cos x \cos 2x}$ $= \frac{3 \sin x \cos 2x + 2 \sin x}{\cos x \cos 2x}$ $= \frac{\sin x \cos 2x + 2 \sin x(\cos 2x + 1)}{\sin x \cos 2x + 2 \sin x(2 \cos^2 x)}$ $= \frac{\cos x \cos 2x}{\sin x \cos 2x + 2(2 \sin x \cos x) \cos x}$ $= \frac{\cos x \cos 2x}{\sin x \cos 2x + 2 \sin 2x \cos x}$ $= \frac{\cos x \cos 2x}{\sin x} + \frac{2 \sin 2x}{\cos 2x}$ $= \tan x + 2 \tan 2x = \text{L.H.S.}$ (b) $\tan x + 2 \tan 2x + 4 \tan 4x + 8 \tan 8x - \cot x = 16$ $\cot x - 4 \cot 4x + 4(\cot 4x - 4 \cot 16x) - \cot x = 16$ $-16 \cot 16x = 16 \implies \cot 16x = -1$ Since $\frac{\pi}{10} < x < \frac{\pi}{8} \implies \frac{8\pi}{5} < 16x < 2\pi$ $16x = 2\pi - \frac{\pi}{4} = \frac{7\pi}{4} \implies x = \frac{7\pi}{64}$	1M 1M 1A 1M 1M 1A	

Solution	Marks	
4. (a) L.H.S. = $\tan^2 \theta + \cot^2 \theta = \frac{\sin^2 \theta}{\cos^2 \theta} + \frac{\cos^2 \theta}{\sin^2 \theta}$ $= \frac{\sin^4 \theta + \cos^4 \theta}{\sin^2 \theta \cos^2 \theta}$ $= \frac{\sin^4 \theta + 2 \sin^2 \theta \cos^2 \theta + \cos^4 \theta - 2 \sin^2 \theta \cos^2 \theta}{\frac{1}{4} \sin^2 2\theta}$ $= \frac{(\sin^2 \theta + \cos^2 \theta)^2 - \frac{1}{2} \sin^2 2\theta}{\frac{1}{4} \sin^2 2\theta}$ $= \frac{1 - \frac{1}{2}(\frac{2}{3})^2}{\frac{1}{4}(\frac{2}{3})^2} = 7 = \text{R.H.S.}$ (b) $\frac{1}{1 + \tan^2 \theta} + \frac{1}{1 + \cot^2 \theta} = \frac{1}{\sec^2 \theta} + \frac{1}{\csc^2 \theta} = \cos^2 \theta + \sin^2 \theta = 1$ $(\tan^2 \theta + \cot^2 \theta)^2 = 7^2 \implies \tan^4 \theta + 2 + \cot^4 \theta = 49 \implies \tan^4 \theta + \cot^4 \theta - 2 = 45$ Expression = $1 - \frac{12}{45} = 1 - \frac{4}{15} = \frac{11}{15}$	1M 1M 1A 1M 1M 1A	

Solution	Marks	
5. (a) $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$ (b)(i) Given $\cos \alpha + \cos \beta + \cos \gamma = 0$ and $\sin \alpha + \sin \beta + \sin \gamma = 0$. $(\cos \alpha + \cos \beta + \cos \gamma)^2 + (\sin \alpha + \sin \beta + \sin \gamma)^2 = 0$ $[\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + 2(\cos \alpha \cos \beta + \cos \beta \cos \gamma + \cos \gamma \cos \alpha)] + [\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma + 2(\sin \alpha \sin \beta + \sin \beta \sin \gamma + \sin \gamma \sin \alpha)] = 0$ $(1+1+1) + 2[(\cos \alpha \cos \beta + \sin \alpha \sin \beta) + (\cos \beta \cos \gamma + \sin \beta \sin \gamma) + (\cos \gamma \cos \alpha + \sin \gamma \sin \alpha)] = 0$ $3 + 2[\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha)] = 0$ $\cos(\alpha - \beta) + \cos(\beta - \gamma) + \cos(\gamma - \alpha) = -\frac{3}{2}$ (b)(ii) $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = 2 \cos(\alpha + \beta) \cos(\alpha - \beta) + \cos 2\gamma$ $\cos \alpha + \cos \beta + \cos \gamma = 0 \implies 2 \cos \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2} = -\cos \gamma \quad \dots\dots(1)$ $\sin \alpha + \sin \beta + \sin \gamma = 0 \implies 2 \sin \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2} = -\sin \gamma \quad \dots\dots(2)$ $(1)^2 + (2)^2 : [2 \cos \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}]^2 + [2 \sin \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}]^2 = (-\cos \gamma)^2 + (-\sin \gamma)^2$ $\implies 4 \cos^2 \frac{\alpha-\beta}{2} \left(\cos^2 \frac{\alpha+\beta}{2} + \sin^2 \frac{\alpha+\beta}{2} \right) = 1$ $2(2 \cos^2 \frac{\alpha-\beta}{2}) = 1$ $2 + 2 \cos(\alpha - \beta) = 1$ $2 \cos(\alpha - \beta) = -1$ $\cos 2\gamma = (-\cos \gamma)^2 - (-\sin \gamma)^2 = (1)^2 - (2)^2$ $= [2 \cos \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}]^2 - [2 \sin \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}]^2$ $= 4 \cos^2 \frac{\alpha-\beta}{2} \left(\cos^2 \frac{\alpha+\beta}{2} - \sin^2 \frac{\alpha+\beta}{2} \right)$ $= (1)(\cos(\alpha + \beta))$ $2 \cos(\alpha + \beta) \cos(\alpha - \beta) + \cos 2\gamma = -\cos(\alpha + \beta) + \cos(\alpha + \beta) = 0$ $\therefore \cos 2\alpha + \cos 2\beta + \cos 2\gamma = 0$ (c) $\sin(\alpha - \phi) \cos(\alpha - \phi) + \sin(\beta - \phi) \cos(\beta - \phi) + \sin(\gamma - \phi) \cos(\gamma - \phi)$ $= \frac{1}{2}[\sin(2\alpha - 2\phi) + \sin(2\beta - 2\phi) + \sin(2\gamma - 2\phi)]$ $= \frac{1}{2}[\cos 2\phi(\sin 2\alpha + \sin 2\beta + \sin 2\gamma) - \sin 2\phi(\cos 2\alpha + \cos 2\beta + \cos 2\gamma)]$ $= \frac{1}{2}[\cos 2\phi(\sin 2\alpha + \sin 2\beta + \sin 2\gamma)]$ $\sin 2\gamma = 2(-\sin \gamma)(-\cos \gamma) = 2(1)(2)$ $= 2[2 \cos \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}][2 \sin \frac{\alpha+\beta}{2} \cos \frac{\alpha-\beta}{2}]$	1A 1M 1A 1M 1M 1A 1M 1M	

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$= [2 \cos \frac{\alpha + \beta}{2} \sin \frac{\alpha + \beta}{2}] [4 \cos^2 \frac{\alpha - \beta}{2}]$ $= \sin(\alpha + \beta)$ $\therefore \sin 2\alpha + \sin 2\beta + \sin 2\gamma = 2 \sin(\alpha + \beta) \cos(\alpha - \beta) + \sin 2\gamma = -\sin(\alpha + \beta) + \sin(\alpha + \beta) = 0$ Value of $\sin(\alpha - \phi) \cos(\alpha - \phi) + \sin(\beta - \phi) \cos(\beta - \phi) + \sin(\gamma - \phi) \cos(\gamma - \phi) = 0$	1A	

Solution	Marks	
6. (a)(i) $\tan 3x = \frac{\tan x + \tan 2x}{1 - \tan x \tan 2x} = \frac{\tan x + \frac{2 \tan x}{1 - \tan^2 x}}{1 - \tan x \cdot \frac{2 \tan x}{1 - \tan^2 x}}$ $= \frac{\tan x(1 - \tan^2 x) + 2 \tan x}{1 - \tan^2 x - 2 \tan^2 x} = \frac{3 \tan x - \tan^3 x}{1 - 3 \tan^2 x}$ (a)(ii) $\tan^2 3\theta = \frac{(3 \tan \theta - \tan^3 \theta)^2}{(1 - 3 \tan^2 \theta)^2} = \frac{9 \tan^2 \theta - 6 \tan^4 \theta + \tan^6 \theta}{1 - 6 \tan^2 \theta + 9 \tan^4 \theta}$ Given $\tan^6 \theta - 33 \tan^4 \theta + 27 \tan^2 \theta - 3 = 0$ $\implies \tan^6 \theta - 6 \tan^4 \theta + 9 \tan^2 \theta = 27 \tan^4 \theta - 18 \tan^2 \theta + 3 = 3(9 \tan^4 \theta - 6 \tan^2 \theta + 1)$ $\tan^2 3\theta = \frac{3(9 \tan^4 \theta - 6 \tan^2 \theta + 1)}{9 \tan^4 \theta - 6 \tan^2 \theta + 1} = 3$ (b) $\tan^2 3\theta = 3 \implies \tan 3\theta = \sqrt{3}$ or $-\sqrt{3}$ Since θ is acute ($0 < \theta < \frac{\pi}{2}$), $0 < 3\theta < \frac{3\pi}{2}$: $3\theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}$ $\theta = \frac{\pi}{9}, \frac{2\pi}{9}, \frac{4\pi}{9}$ (c) Product $= \tan \frac{\pi}{9} \tan \frac{2\pi}{9} \tan \frac{4\pi}{9}$ $= \frac{\sin \frac{\pi}{9} \sin \frac{2\pi}{9} \sin \frac{4\pi}{9}}{\cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}}$ $= \frac{8 \sin \frac{\pi}{9} \sin \frac{\pi}{9} \sin \frac{2\pi}{9} \sin \frac{4\pi}{9}}{8 \sin \frac{\pi}{9} \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}}$ $= \frac{4 \sin(\pi - \frac{\pi}{9}) \sin \frac{\pi}{9} (\cos \frac{2\pi}{9} - \cos \frac{6\pi}{9})}{\sin \frac{8\pi}{9}}$ $= 4 \sin \frac{\pi}{9} (\cos \frac{2\pi}{9} - \frac{-1}{2})$ $= 4 \sin \frac{\pi}{9} \cos \frac{2\pi}{9} + 2 \sin \frac{\pi}{9}$ $= 2(\sin \frac{3\pi}{9} + \sin \frac{-\pi}{9}) + 2 \sin \frac{\pi}{9}$ $= \sqrt{3} - 2 \sin \frac{\pi}{9} + 2 \sin \frac{\pi}{9}$ $= \sqrt{3}$	1M 1A 1M 1M 1A 1M 1M 1A	

Solution	Marks	
7. (a) Put $n = 3$ and $\theta = \frac{\pi}{7}$: $\frac{\sin(7 \cdot \frac{\pi}{7})}{\sin \frac{\pi}{7}} = 1 + 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right)$ Since $\sin \pi = 0$, $0 = 1 + 2 \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right)$ $\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} = -\frac{1}{2}$ (b)(i) $\cos 3x = \cos(2x + x) = \cos 2x \cos x - \sin 2x \sin x$ $= (2 \cos^2 x - 1) \cos x - (2 \sin x \cos x) \sin x$ $= 2 \cos^3 x - \cos x - 2 \cos x (1 - \cos^2 x) = 4 \cos^3 x - 3 \cos x$ (b)(ii) $\cos 4\theta = \cos(2 \cdot 2\theta) = 2 \cos^2 2\theta - 1 = 2(2 \cos^2 \theta - 1)^2 - 1 = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$ $\cos 3\theta = \cos 4\theta \implies 4 \cos^3 \theta - 3 \cos \theta = 8 \cos^4 \theta - 8 \cos^2 \theta + 1$ $8 \cos^4 \theta - 4 \cos^3 \theta - 8 \cos^2 \theta + 3 \cos \theta + 1 = 0$ $(\cos \theta - 1)(8 \cos^3 \theta + 4 \cos^2 \theta - 4 \cos \theta - 1) = 0$ For $\theta = \frac{2\pi}{7}$, $\cos \frac{2\pi}{7} \neq 1$, so $8 \cos^3 \theta + 4 \cos^2 \theta - 4 \cos \theta - 1 = 0$ (b)(iii) $\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7} = \cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos(\pi - \frac{\pi}{7})$ $= -\cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}$ $= \frac{-\sin \frac{\pi}{7} \cos \frac{\pi}{7} \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}}{\sin \frac{\pi}{7}}$ $= \frac{-\frac{1}{8} \sin \frac{8\pi}{7}}{\sin \frac{\pi}{7}}$ $= \frac{1}{8}$ (c)(i) $\cos^2 \frac{2\pi}{7} + \cos^2 \frac{4\pi}{7} + \cos^2 \frac{6\pi}{7} = \frac{1 + \cos \frac{4\pi}{7}}{2} + \frac{1 + \cos \frac{8\pi}{7}}{2} + \frac{1 + \cos \frac{12\pi}{7}}{2}$ $= \frac{3}{2} + \frac{1}{2} \left(\cos \frac{4\pi}{7} + \cos \frac{8\pi}{7} + \cos \frac{12\pi}{7} \right)$ $= \frac{3}{2} + \frac{1}{2} \left[\cos(2\pi - \frac{4\pi}{7}) + \cos(2\pi - \frac{8\pi}{7}) + \cos(2\pi - \frac{12\pi}{7}) \right]$ $= \frac{3}{2} + \frac{1}{2} \left(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7} \right)$ $= \frac{3}{2} + \frac{1}{2} \left(-\frac{1}{2} \right) = \frac{5}{4}$ (c)(ii) $\sec \frac{2\pi}{7} + \sec \frac{4\pi}{7} + \sec \frac{6\pi}{7} = \frac{\cos \frac{4\pi}{7} \cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} \cos \frac{6\pi}{7} + \cos \frac{2\pi}{7} \cos \frac{4\pi}{7}}{\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}}$ $= \frac{\frac{1}{2} \left[(\cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{6\pi}{7})^2 - (\cos^2 \frac{2\pi}{7} + \cos^2 \frac{4\pi}{7} + \cos^2 \frac{6\pi}{7}) \right]}{\cos \frac{2\pi}{7} \cos \frac{4\pi}{7} \cos \frac{6\pi}{7}}$ $= \frac{\frac{1}{2} [(-\frac{1}{2})^2 - \frac{5}{4}]}{\frac{1}{8}} = -4$	1M 1A 1M 1A 1M 1M 1A 1M 1A 1M 1A	

Solution	Marks	
SECTION B 8. (a) Let $P(n)$ be the proposition $\sum_{k=1}^n \frac{1}{2^k} \tan\left(\frac{\theta}{2^k}\right) = \frac{1}{2^n} \cot\left(\frac{\theta}{2^n}\right) - \cot \theta$. For $n = 1$: L.H.S. $= \frac{1}{2} \tan\left(\frac{\theta}{2}\right)$ R.H.S. $= \frac{1}{2} \cot\left(\frac{\theta}{2}\right) - \cot \theta = \frac{1}{2} \cot\left(\frac{\theta}{2}\right) - \frac{\cot^2(\theta/2) - 1}{2 \cot(\theta/2)} = \frac{1}{2 \cot(\theta/2)} = \frac{1}{2} \tan\left(\frac{\theta}{2}\right)$ $\therefore$ L.H.S. = R.H.S., $P(1)$ is true. Assume $P(m)$ is true for some positive integer m, i.e., $\sum_{k=1}^m \frac{1}{2^k} \tan\left(\frac{\theta}{2^k}\right) = \frac{1}{2^m} \cot\left(\frac{\theta}{2^m}\right) - \cot \theta.$ For $n = m + 1$: $\begin{aligned} \sum_{k=1}^{m+1} \frac{1}{2^k} \tan\left(\frac{\theta}{2^k}\right) &= \frac{1}{2^m} \cot\left(\frac{\theta}{2^m}\right) - \cot \theta + \frac{1}{2^{m+1}} \tan\left(\frac{\theta}{2^{m+1}}\right) \\ &= \frac{1}{2^{m+1}} \left[2 \cot\left(\frac{\theta}{2^m}\right) + \tan\left(\frac{\theta}{2^{m+1}}\right) \right] - \cot \theta \\ &= \frac{1}{2^{m+1}} \left[2 \cot\left(2 \cdot \frac{\theta}{2^{m+1}}\right) + \tan\left(\frac{\theta}{2^{m+1}}\right) \right] - \cot \theta \\ &= \frac{1}{2^{m+1}} \left[2 \cdot \frac{1 - \tan^2\left(\frac{\theta}{2^{m+1}}\right)}{2 \tan\left(\frac{\theta}{2^{m+1}}\right)} + \tan\left(\frac{\theta}{2^{m+1}}\right) \right] - \cot \theta \\ &= \frac{1}{2^{m+1}} \cot\left(\frac{\theta}{2^{m+1}}\right) - \cot \theta \end{aligned}$ $\therefore P(m + 1)$ is true. By mathematical induction, $P(n)$ is true for all positive integers n. (b) Let $\theta = \frac{\pi}{3}$. $\begin{aligned} \sum_{k=5}^{10} \frac{1}{2^k} \tan\left(\frac{\pi}{2^k}\right) &= \sum_{k=1}^{10} \frac{1}{2^k} \tan\left(\frac{\pi}{2^k}\right) - \sum_{k=1}^4 \frac{1}{2^k} \tan\left(\frac{\pi}{2^k}\right) \\ &= \left[\frac{1}{1024} \cot\left(\frac{\pi}{3072}\right) - \cot \frac{\pi}{3} \right] - \left[\frac{1}{16} \cot\left(\frac{\pi}{48}\right) - \cot \frac{\pi}{3} \right] \\ &= \frac{1}{1024} \cot\left(\frac{\pi}{3072}\right) - \frac{1}{16} \cot\left(\frac{\pi}{48}\right) \end{aligned}$	1M 1M 1M 1A 1M 1A	

Solution	Marks	
9. (a) R.H.S. = $\frac{\cos 2\theta}{1 + \sin 2\theta} = \frac{\cos^2 \theta - \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta}$ $= \frac{(\cos \theta - \sin \theta)(\cos \theta + \sin \theta)}{(\cos \theta + \sin \theta)^2} = \frac{\cos \theta - \sin \theta}{\cos \theta + \sin \theta}$ $= \frac{1 - \tan \theta}{1 + \tan \theta} = \frac{\tan(\frac{\pi}{4}) - \tan \theta}{1 + \tan(\frac{\pi}{4}) \tan \theta} = \tan\left(\frac{\pi}{4} - \theta\right) = \text{L.H.S.}$ (b) By Binomial Theorem, $\sum_{r=0}^{n+1} C_r^{n+1} (-1)^r (\sin 2\phi)^r (1 + \sin 2\phi)^{n+1-r} = [(1 + \sin 2\phi) - \sin 2\phi]^{n+1} = 1^{n+1} = 1.$ Splitting the sum into $r = 0$, $r = 1 \dots n$, and $r = n + 1$: $(1 + \sin 2\phi)^{n+1} + \sum_{r=1}^n C_r^{n+1} (-1)^r (\sin 2\phi)^r (1 + \sin 2\phi)^{n+1-r} + (-1)^{n+1} (\sin 2\phi)^{n+1} = 1$ $\therefore \sum_{r=1}^n C_r^{n+1} (-1)^r (\sin 2\phi)^r (1 + \sin 2\phi)^{n+1-r} = 1 - (1 + \sin 2\phi)^{n+1} - (-\sin 2\phi)^{n+1}$ (c) $\sum_{r=1}^3 C_r^4 \left(-\frac{1}{2}\right)^r (\sin 4\phi)^r (\cos 2\phi)^{4-2r} \left[\tan\left(\frac{\pi}{4} - \phi\right)\right]^{r-4}$ $= \sum_{r=1}^3 C_r^4 \left(-\frac{1}{2}\right)^r (2 \sin 2\phi \cos 2\phi)^r (\cos 2\phi)^{4-2r} \left(\frac{\cos 2\phi}{1 + \sin 2\phi}\right)^{r-4}$ $= \sum_{r=1}^3 C_r^4 (-\sin 2\phi)^r (\cos 2\phi)^{4-r} (\cos 2\phi)^{r-4} \left(\frac{1}{1 + \sin 2\phi}\right)^{r-4}$ $= \sum_{r=1}^3 C_r^4 (-\sin 2\phi)^r (1 + \sin 2\phi)^{4-r}$ $= (-\sin 2\phi + 1 + \sin 2\phi)^4 - (-\sin 2\phi)^4 - (1 + \sin 2\phi)^4$ $= 1 - \left(-\sin \frac{\pi}{4}\right)^4 - \left(1 + \sin \frac{\pi}{4}\right)^4$ $= 1 - \left(-\frac{\sqrt{2}}{2}\right)^4 - \left(1 + \frac{\sqrt{2}}{2}\right)^4 = -\frac{7}{2} - 3\sqrt{2}$	1M 1A 1M 1A 1M 1A	

Solution	Marks	
10. (a) R.H.S. = $\sin \theta \left[\frac{1}{6(1 - \cos \theta)} + \frac{1}{2(1 + \cos \theta)} - \frac{1}{3(2 + \cos \theta)} \right]$ $= \sin \theta \left[\frac{(1 + \cos \theta)(2 + \cos \theta) + 3(1 - \cos \theta)(2 + \cos \theta) - 2(1 - \cos^2 \theta)}{6(1 - \cos^2 \theta)(2 + \cos \theta)} \right]$ $= \sin \theta \left[\frac{(2 + 3 \cos \theta + \cos^2 \theta) + (6 - 3 \cos \theta - 3 \cos^2 \theta) - (2 - 2 \cos^2 \theta)}{6 \sin^2 \theta (2 + \cos \theta)} \right]$ $= \sin \theta \left[\frac{6}{6 \sin^2 \theta (2 + \cos \theta)} \right] = \frac{1}{\sin \theta (2 + \cos \theta)} = \text{L.H.S.}$ (b) $I = \int \frac{\sin \theta d\theta}{6(1 - \cos \theta)} + \int \frac{\sin \theta d\theta}{2(1 + \cos \theta)} - \int \frac{\sin \theta d\theta}{3(2 + \cos \theta)}$ $= \frac{1}{6} \ln(1 - \cos \theta) - \frac{1}{2} \ln(1 + \cos \theta) + \frac{1}{3} \ln(2 + \cos \theta) + C_1$ Using $1 - \cos \theta = 2 \sin^2 \frac{\theta}{2}$ and $1 + \cos \theta = 2 \cos^2 \frac{\theta}{2}$ and $2 + \cos \theta = 3 - 2 \sin^2 \frac{\theta}{2}$: $= \frac{1}{6} \ln \left(2 \sin^2 \frac{\theta}{2} \right) - \frac{1}{2} \ln \left(2 \cos^2 \frac{\theta}{2} \right) + \frac{1}{3} \ln(2 + \cos \theta) + C_1$ $= \frac{1}{3} \ln \left(\sin \frac{\theta}{2} \right) - \ln \left(\cos \frac{\theta}{2} \right) + \frac{1}{3} \ln(2 + \cos \theta) + C_2$ $= \frac{1}{3} \ln \left(\frac{\sin \frac{\theta}{2} (2 + \cos \theta)}{\cos^3 \frac{\theta}{2}} \right) + C_2$ $= \frac{1}{3} \ln \left(\tan \frac{\theta}{2} \cdot \frac{2 + (1 - 2 \sin^2 \frac{\theta}{2})}{\cos^2 \frac{\theta}{2}} \right) + C_2 = \frac{1}{3} \ln \left(\tan \frac{\theta}{2} \cdot (3 \sec^2 \frac{\theta}{2} - 2 \tan^2 \frac{\theta}{2}) \right) + C_2$ $= \frac{1}{3} \ln \left(\tan \frac{\theta}{2} (3(1 + \tan^2 \frac{\theta}{2}) - 2 \tan^2 \frac{\theta}{2}) \right) + C_2$ $= \frac{1}{3} \ln \left(3 \tan \frac{\theta}{2} + \tan^3 \frac{\theta}{2} \right) + C$	1M 1A 1M 1M 1M 1A	

Solution	Marks	
11. (a) $f(x) = \sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}} = \sqrt{\frac{\cos^2 x + \sin^2 x - 2 \sin x \cos x}{\cos^2 x + \sin^2 x + 2 \sin x \cos x}} = \sqrt{\frac{(\cos x - \sin x)^2}{(\cos x + \sin x)^2}}$ For $0 < x < \frac{\pi}{4}$, $\cos x > \sin x > 0$, so $\cos x - \sin x > 0$ and $\cos x + \sin x > 0$. $\therefore f(x) = \frac{\cos x - \sin x}{\cos x + \sin x}$. (b) $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ $= \lim_{h \rightarrow 0} \frac{1}{h} \left[\frac{\cos(x+h) - \sin(x+h)}{\cos(x+h) + \sin(x+h)} - \frac{\cos x - \sin x}{\cos x + \sin x} \right]$ $= \lim_{h \rightarrow 0} \frac{[\cos(x+h) - \sin(x+h)](\cos x + \sin x) - (\cos x - \sin x)[\cos(x+h) + \sin(x+h)]}{h(\cos(x+h) + \sin(x+h))(\cos x + \sin x)}$ $= \lim_{h \rightarrow 0} \frac{2 \sin x \cos(x+h) - 2 \cos x \sin(x+h)}{h(\cos(x+h) + \sin(x+h))(\cos x + \sin x)}$ $= \lim_{h \rightarrow 0} \frac{-2 \sin((x+h) - x)}{h(\cos(x+h) + \sin(x+h))(\cos x + \sin x)}$ $= \lim_{h \rightarrow 0} \left(\frac{\sin h}{h} \right) \cdot \frac{-2}{(\cos(x+h) + \sin(x+h))(\cos x + \sin x)}$ $= 1 \cdot \frac{-2}{(\cos x + \sin x)^2} = \frac{-2}{\cos^2 x + \sin^2 x + 2 \sin x \cos x} = \frac{-2}{1 + \sin 2x}$	1M 1A 1M 1M 1M 1A	

Solution	Marks	
12. (a) $\vec{u} \times \vec{v} = (\cos \theta \vec{a} + \sin \theta \vec{b}) \times (\sin \theta \vec{a} - \cos \theta \vec{b})$ $= -\cos^2 \theta (\vec{a} \times \vec{b}) + \sin^2 \theta (\vec{b} \times \vec{a}) = -\cos^2 \theta (\vec{a} \times \vec{b}) - \sin^2 \theta (\vec{a} \times \vec{b})$ $= -(\cos^2 \theta + \sin^2 \theta) (\vec{a} \times \vec{b}) = -(\vec{a} \times \vec{b}).$ (b)(i) Let $\vec{OQ} = \vec{a} + 2\vec{b}$. $\vec{PQ} \cdot \vec{OQ} = 0 \cdot \vec{a} + 0 \cdot 2\vec{b} + (\vec{a} \times \vec{b}) \cdot 0 = 0$ $\therefore \vec{PQ} \perp \vec{OQ}$ $d = \vec{PQ}$ $= \vec{a} \times \vec{b} = \vec{a} \vec{b} \sin \frac{\pi}{3}$ $= (1)(1) \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{2}.$ (b)(ii) Since $d = \frac{\sqrt{3}}{2}$, which is a constant independent of θ. (c)(i) $\vec{a} \cdot \vec{b} = \vec{a} \vec{b} \cos \frac{\pi}{3} = \frac{1}{2}.$ $\vec{u} \cdot \vec{v} = \sin \theta \cos \theta - \cos^2 \theta (\vec{a} \cdot \vec{b}) + \sin^2 \theta (\vec{a} \cdot \vec{b}) - \sin \theta \cos \theta = -\frac{1}{2} \cos 2\theta.$ $\vec{u} ^2 = \cos^2 \theta + \sin^2 \theta + 2 \sin \theta \cos \theta (\vec{a} \cdot \vec{b}) = 1 + \frac{1}{2} \sin 2\theta.$ $\vec{v} ^2 = \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta (\vec{a} \cdot \vec{b}) = 1 - \frac{1}{2} \sin 2\theta.$ $\vec{w} ^2 = \vec{u} + \tan \theta \vec{v} ^2 = \vec{u} ^2 + 2 \tan \theta (\vec{u} \cdot \vec{v}) + \tan^2 \theta \vec{v} ^2$ $= (1 + \frac{1}{2} \sin 2\theta) + 2 \tan \theta (-\frac{1}{2} \cos 2\theta) + \tan^2 \theta (1 - \frac{1}{2} \sin 2\theta)$ $= 1 + \sin \theta \cos \theta - \tan \theta (\cos^2 \theta - \sin^2 \theta) + \tan^2 \theta - \tan^2 \theta \sin \theta \cos \theta = 1 + \tan^2 \theta = \sec^2 \theta.$ (c)(ii) Let $g(\theta) = \vec{w} ^2 = \sec^2 \theta.$ $\frac{d}{d\theta} g(\theta) = 2 \sec^2 \theta \tan \theta.$ For $0 \leq \theta < \frac{\pi}{2}$, $2 \sec^2 \theta \tan \theta \geq 0$, and equals 0 when $\theta = 0$. Minimum value of $\vec{w} ^2$ is $g(0) = \sec^2 0 = 1.$	1M 1A 1M 1M 1A 1A 1M 1A 1M 1A	

Solution	Marks	
13. (a) $M = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ $M^2 = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix}$ $= \begin{pmatrix} \cos^2 \theta - \sin^2 \theta & 2 \sin \theta \cos \theta \\ -2 \sin \theta \cos \theta & \cos^2 \theta - \sin^2 \theta \end{pmatrix} = \begin{pmatrix} \cos 2\theta & \sin 2\theta \\ -\sin 2\theta & \cos 2\theta \end{pmatrix}.$ $M^2 - \sqrt{3}M + I = \begin{pmatrix} \cos 2\theta - \sqrt{3} \cos \theta + 1 & \sin 2\theta - \sqrt{3} \sin \theta \\ -\sin 2\theta + \sqrt{3} \sin \theta & \cos 2\theta - \sqrt{3} \cos \theta + 1 \end{pmatrix}.$ $\det(M^2 - \sqrt{3}M + I) = (\cos 2\theta - \sqrt{3} \cos \theta + 1)^2 + (\sin 2\theta - \sqrt{3} \sin \theta)^2 = 0$ $(\cos 2\theta - \sqrt{3} \cos \theta + 1)^2 + (\sin 2\theta - \sqrt{3} \sin \theta)^2$ $= \cos^2 2\theta + 3 \cos^2 \theta + 1 - 2\sqrt{3} \cos 2\theta \cos \theta + 2 \cos 2\theta - 2\sqrt{3} \cos \theta + \sin^2 2\theta - 2\sqrt{3} \sin 2\theta \sin \theta + 3 \sin^2 \theta$ $= 1 + 3 + 1 - 2\sqrt{3}(\cos 2\theta \cos \theta + \sin 2\theta \sin \theta) + 2 \cos 2\theta - 2\sqrt{3} \cos \theta$ $= 5 - 2\sqrt{3} \cos \theta + 2 \cos 2\theta - 2\sqrt{3} \cos \theta = 2 \cos 2\theta - 4\sqrt{3} \cos \theta + 5 = 0.$ (b) $2(2 \cos^2 \theta - 1) - 4\sqrt{3} \cos \theta + 5 = 0 \implies 4 \cos^2 \theta - 4\sqrt{3} \cos \theta + 3 = 0.$ $(2 \cos \theta - \sqrt{3})^2 = 0 \implies \cos \theta = \frac{\sqrt{3}}{2}.$ Since $0 < \theta < \frac{\pi}{2}$, $\theta = \frac{\pi}{6}$. (c) $M = \begin{pmatrix} \cos \frac{\pi}{6} & \sin \frac{\pi}{6} \\ -\sin \frac{\pi}{6} & \cos \frac{\pi}{6} \end{pmatrix} = \begin{pmatrix} \frac{\sqrt{3}}{2} & \frac{1}{2} \\ -\frac{1}{2} & \frac{\sqrt{3}}{2} \end{pmatrix}$ $M^2 = \begin{pmatrix} \cos \frac{\pi}{3} & \sin \frac{\pi}{3} \\ -\sin \frac{\pi}{3} & \cos \frac{\pi}{3} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix} = \sqrt{3}M - I$ $M^4 = M^2 M^2 = (\sqrt{3}M - I)^2 = 3M^2 - 2\sqrt{3}M + I = 3(\sqrt{3}M - I) - 2\sqrt{3}M + I = \sqrt{3}M - 2I$ $M^6 = M^4 M^2 = (\sqrt{3}M - 2I)(\sqrt{3}M - I) = 3M^2 - 3\sqrt{3}M + 2I$ $= 3(\sqrt{3}M - I) - 3\sqrt{3}M + 2I = -I$ $M^{2026} = (M^6)^{337} M^4 = (-I)^{337} (\sqrt{3}M - 2I) = (-\sqrt{3}M + 2I)$ $\therefore a_{2026} = \sqrt{3}.$	1M 1M 1A 1M 1M 1A 1A 1A	